

Project Overview

1 Introduction

This is a simple script used to obtain an airfoil optimized for maximum aerodynamic efficiency (i.e., L/D), at some given inputs.

From the user, the script requires:

1. Reynolds number
2. Range of Angles of Attack to analyze
3. Airfoil parametrization mode
4. Bounds for airfoil parametrization

`airfoilOpt.py` is the main script for running the optimization. `airfoilTools.py` contains definitions for the airfoil parametrization, analysis, and post-processing. `monitor.py` can be run after the optimization is complete to visualize the evolution of L/D throughout the run.

The optimization workflow happens through the `differential_evolution` optimizer from **SciPy**. The function makes its evaluation by calling **XFOIL**, processes its outputs, and returns the combination of parameters that produce the best L/D within the specified airfoil parametrization bounds.

2 Uncertainties

2.1 XFOIL Verification and Validation

XFOIL is a well-validated code that has been in use in the aerospace industry for over 20 years. There have been many studies regarding its performance ([4], [3], [1]) and the original documentation by Drela [2]. To summarize their findings, XFOIL is able to reliably predict C_l in the linear region of the lift curve. When you get to the C_{Lmax} region, inaccuracies occur. Airfoils experience separation near the region of peak C_l , and XFOIL is unable to accurately handle this phenomenon. This also means that stall prediction and values near stall are also unreliable. However, for the purposes of this project as a learning experience, these inaccuracies can be accepted. The main benefit of XFOIL here, is its speed. The program can test around 2000 airfoils in a few minutes, depending on the number of cores used, and this speed makes it very useful for an optimization study.

2.2 Code Verification

Verification of the code has not been done yet. This is the next stage of the project

References

- [1] James G. Coder and Mark D. Maughmer. “Comparisons of Theoretical Methods for Predicting Airfoil Aerodynamic Characteristics”. In: *Journal of Aircraft* 51.1 (Jan. 2014), pp. 183–191. DOI: [10.2514/1.c032232](https://doi.org/10.2514/1.c032232).
- [2] Mark Drela. “XFOIL: an analysis and design system for Low Reynolds number airfoils”. In: *Low Reynolds number aerodynamics*. Springer, 1989, pp. 1–12. DOI: [10.1007/978-3-642-84010-4_1](https://doi.org/10.1007/978-3-642-84010-4_1).

- [3] Cody Lafountain, Kelly Cohen, and Shaaban Abdallah. “Use of XFOIL in design of camber-controlled morphing UAVs”. In: *Computer Applications in Engineering Education* 20.4 (Oct. 2012), pp. 673–680. DOI: [10.1002/cae.20437](https://doi.org/10.1002/cae.20437).
- [4] J. Morgado et al. “XFOIL vs CFD performance predictions for high lift low Reynolds number airfoils”. In: *Aerospace Science and Technology* 52 (2016), pp. 207–214. DOI: [10.1016/j.ast.2016.02.031](https://doi.org/10.1016/j.ast.2016.02.031).